

Analogue Components for IoT Applications

Numerous analogue components and chips are introduced to the market every year, but what is influencing their evolution? Read on to find out the latest technologies in this field, the influence of the Internet of Things (IoT) paradigm on analogue world and some interesting applications powered by analogue components

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Thirty years ago, almost all the technology surrounding us was analogue. However, in the past decade we have witnessed a massive shift from analogue to digital electronics. The digital platform offers amazing opportunities for innovation, and is thus promptly taking over several aspects of modern life.

Will the digital world be able to completely take over analogue technologies and devices? Don't think so. What we found is that analogue is still well-established, and innovative technologies and products are continuing to emerge in this sector. What is driving this 'old' technology?

Influence of IoT

The IoT is perhaps one of the most discussed technology topics these days. It provides a huge opportunity for us to leverage the Internet infrastructure to put technology to better use. Its biggest enabler is the ability to connect the real world with the digital world, thus enabling people and machines to know the status of hundreds of thousands of objects simultaneously.

Analogue components are the key to deliver this connection between the real and digital worlds.

While back-end of the IoT is communication and connectivity forming the digital side, the front-end generally consists of sensors (temperature, pressure, motion, tilt, acceleration, shock, proximity, gas, sound, pH, infrared, etc), actuators, power management modules (linear regulator, converter,



STC3115 battery monitoring IC by STMicroelectronics

battery charger), signal acquisition modules, amplifiers, signal processors, microcontrollers and integrated chips; this is where analogue comes into play.

New component design elements

Form factor and power are major considerations for smart products, and the industry is getting geared up for making analogue devices suitable for IoT segment. However, there are some cases where the size and/or power are not of high relevance, like for the smart-grid applications. Some of the trends driving the analogue industry are discussed in detail below.

Wearable medical devices and low-power components. Many devices, particularly portable and wearable electronics, cannot afford battery-hungry processors and components—even those previously considered as low-power.

This has led to the development of low and ultra-low power devices that can operate for days and months on a single battery, or without the need for recharging. For example, OPA2652 operational amplifiers (op-amps) by Texas Instruments (TI) work at an input current as low as 300nA. On a coin cell it operates for several months together.

Consider a patient-monitoring system. If the instrument is easily drained of all power and has to be charged every few hours, proper monitoring cannot be attained. Moreover, the whole charging process is troubling for the user too.

Benefits of digital in analogue world. The major trick to reduce power is to use low voltage. We want a technology that can be used in analogue devices, operates in low voltages, and also integrates to high voltages. This is where digital was merged with analogue.